

American Economic Association

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Source: *The American Economic Review*, Vol. 77, No. 5 (Dec., 1987), pp. 856-874

Published by: [American Economic Association](#)

Stable URL: <http://www.jstor.org/stable/1810213>

Accessed: 28/06/2014 18:59

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An Equilibrium Model with Involuntary Unemployment at Flexible, Competitive Prices and Wages

By JOHN ROBERTS*

This paper presents a general equilibrium model in which all prices and quantities transacted are explicitly chosen by economic agents: there is no Walrasian auctioneer. Multiple equilibria occur with prices and wages taking their Walrasian values. Equilibrium quantities may also be Walrasian, or they may involve some price-taking workers being rationed in selling labor. This involuntary unemployment results from self-confirming expectations of inadequate effective demand, as in some interpretations of J. M. Keynes' ideas.

The purpose of this paper is to attempt to reconcile the notion of involuntary unemployment with the hypothesis of equilibrium by constructing a closed, reasonably complete economic model which admits such unemployment as an equilibrium phenomenon.

The model in fact generates multiple equilibria which involve Walrasian, perfectly competitive prices but differing levels of economic activity. Equilibrium with full employment exists, with all agents transacting their Walrasian quantities. Simultaneously there are also equilibria at these same prices and wages in which markets fail to clear. In particular, some price- and wage-taking workers are rationed in their labor market transactions and are unable to sell as much of their labor as they desire at the given wage.¹ This involuntary unemployment arises

despite the model's incorporating markets for all commodities. Further, the levels of all nominal prices and wages are endogenously determined, and in fact they are treated as choice variables of regular maximizing economic agents, as are the amounts of each good bought and sold by each agent. The agents are fully informed about one another's preferences, endowments, and production possibilities, and there is no uncertainty about any of these. Nor are markets in any sense physically separated. The agents also understand the institutions of price and quantity determination, and when making their choices they have full knowledge of any choices that have been made previously. As well, in equilibrium the agents are each acting optimally in every eventuality (not just those that would arise under some putative equilibrium behavior) while correctly forecasting both one another's choices of prices and quantities and the full implications of adopting any available course of action. In particular, no agent who is in a position to influence some prices or wages is ever mistaken about the effect of changing these, and in equilibrium no such agent finds it worthwhile, for example, to reduce wages in the face of involuntary unemployment.

It is clear that a model with these properties must depart from orthodoxy in some significant fashion. Although we assume a special structure of preferences, endowments, and technologies, under which no potential customer of a firm is also one of its potential employees, the key is in the model-

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¹Taking this as a definition of involuntary unemployment seems to fit with much of the discussion in Chap. 2 of *The General Theory* (J. M. Keynes, 1936).

ing of the processes determining prices and individual transactions.

Neither of these processes is explicitly modeled in detail in standard equilibrium models in the tradition of Cournot and Walras. In these models, no economic agent actually sets prices. Instead, prices somehow emerge "from the market" and take whatever values are required to equate aggregate offers to buy and sell. Then, given such prices, the orders and offers that an individual has announced to the market are somehow filled, although the actual transactions that the individual makes with other agents are not generally modeled. Moreover, the results of agents' adopting nonequilibrium forms of behavior or of the implicit price adjustment and order-filling mechanisms' failing to operate are typically not specified. Thus, these models provide no formal basis for "disequilibrium" analysis.

Clearly, involuntary unemployment cannot exist in equilibrium in such models, because equilibrium involves market clearing in its very definition. If desires to buy and sell do not match, then we do not have an equilibrium and, if we assume that equilibrium will obtain (which is all we can do if we want to use these models), something will have to change. However, exactly what changes and how this occurs is outside the purview of the model; indeed, formally we cannot even say if any of the agents in the model have both the ability and the incentive to effect the requisite changes.

The model offered here is explicit about the operation of these processes, both in and out of equilibrium. Specifically, in the basic model considered in Section I, firms are treated as announcing prices for the goods they can produce and for the inputs they can use.² This means that there may be different prices being quoted by different firms for the same good. Knowing the announced prices, consumers place output purchase orders and input supply offers with specific firms. Actual quantities transacted are then de-

termined by the firms' decisions of how much of these orders and offers to accept.³

Thus, the institutions for price and quantity determination are described by specifying which agents can act at various points, what options are open to them, what information they have when taking their actions, and what outcomes result from each set of action choices. Given preferences, endowments, and production possibilities, specifying a set of institutions in this way yields a game in extensive form. Within such a game, a (pure) strategy for an agent is a specification of the action the person will take in every possible circumstance in which he or she might have to act, not just in those that arise under some putative equilibrium mode of behavior. A well-defined outcome results from every assignment of strategies to agents, and so, given conjectures about how others are acting, each agent can determine the implications of adopting any course of behavior.

To solve this game, we search for subgame perfect equilibria (Reinhard Selten, 1975). These consist of a strategy for each agent with the property that, for each agent and for each situation in which the agent must act, adhering to the strategy and adopting the behavior specified by it is optimal for the agent, given that the other agents' current and future actions will be governed by their strategies. Equilibrium thus means first that at each decision point, each agent correctly forecasts the actions that others are currently taking and the responses from the other agents that would be elicited by the various choices the agent might make. Further, given these correct expectations, the agent chooses at each point the course of action that is optimal from that point forward. Thus, in an equilibrium with involuntary unemployment in this model, every agent correctly perceives that, taking account of the actions and reactions of the other agents, no unilateral change in behavior will benefit him or her. In par-

²There is only one good which does not enter production functions, and it is used as numeraire, with its price set at unity.

³Variants of this basic model, considered in Section II, have labor market transactions completed before output orders are placed and allow workers to set wages.

ticular, the various agents who can influence prices and wages correctly forecast that they individually will not gain by changing their prices, and the unemployed workers correctly perceive that there is nothing that anyone of them individually can do that will lead to gainful employment.

There is, of course, a huge literature that aims at generating macroeconomic inefficiency and unemployment, and it is impossible to give any satisfactory account of it here. However, the present analysis connects most clearly to a relatively few, very prominent strands of this work, and it is worthwhile mentioning these.

First is the idea of Keynesian effective demand failures based on self-confirming conjectures (see Robert Clower, 1965, and Axel Leijonhufvud, 1968): firms are unwilling to increase hiring because each forecasts that demand will be too weak to justify its increasing output, and the resultant low level of workers' incomes generates the weak demand that makes this conjecture correct. However, if all firms increased hiring together, the additional income generated could result in enough extra demand to justify the hiring. The models offered here are meant to capture formally just such notions.

Of course, these ideas have been formalized previously by Robert Barro and Herschel Grossman (1971), Jean-Pascal Benassy (1973), Jacques Drèze (1975), and others (see Benassy, 1982, for references), and aspects of their analyses reappear in the present model. Particularly important is the role of (perceived) quantity constraints and rationing in embodying the basic idea that demand constrains employment. However, in these earlier models at least some prices and wages are assumed to be fixed at levels that are not market clearing, and the models offer no basis for analysis of the opportunities for changing prices or of the incentives to do so. The present model specifically addresses these issues. Moreover, these treatments often assume that there is some unmodeled mechanism at work that ensures the maximum level of transactions consistent with voluntary exchange at the fixed prices and feasibility. Here, the determination of quantities is made explicit.

In this context, Takatoshi Ito (1979) has shown that the possibility of stochastic rationing can lead to equilibria with involuntary unemployment even when prices are exogenously fixed at their Walrasian levels. As in the Clower-Leijonhufvud interpretation of J. M. Keynes, these low-level equilibria are the result of self-confirming pessimism about demand. However, the equilibrium concept that Ito uses does not allow agents to recognize how the probability of their being rationed might depend on their announced orders and offers. Thus, unless there is a non-atomic continuum of agents, individual behavior does not appear to be fully rational in the very strong sense employed in the present paper.

A second important line of work is that dealing with macroeconomic "coordination failures" arising in non-Walrasian market settings involving search (for example, Peter Diamond, 1982; Alan Drazen, 1986) or imperfect competition (Martin Weitzman, 1982; Oliver Hart, 1982; Walter P. Heller, 1986; John Roberts, 1987; Russell Cooper and Andrew John, 1985). This literature displays the possibility of multiple equilibria and inefficiently low equilibrium levels of employment. It does not, however, generate involuntary unemployment in the sense that workers facing given prices and wages are off their supply curves in equilibrium. Instead, workers in these models transact the quantities they desire, but these are inefficiently low because imperfectly competitive restrictions on output depress labor demand and wages or because workers are monopsonistically restricting labor supply.⁴ In contrast, the unemployment that arises in the present model is involuntary in precisely the sense indicated above.

A particularly striking example of coordination failures is provided by John Bryant (1983). He identifies a continuum of "rational expectations" equilibria in an economy in which different agents produce per-

⁴The Weitzman paper does obtain a sort of involuntary unemployment, but only by (in effect) fixing real wages outside the model.

fectly complementary goods. All but one of these equilibria involve inefficiently low levels of activity; the exception is one of the Walrasian allocations for the economy. The present model generates just such a continuum, and although technological complementarities play no role here, the fundamental source of the multiplicity is the same as in Bryant's work: even given prices, different agents' optimal actions are made interdependent by recognition of feasibility constraints. The inclusion of a well-specified, reasonably realistic set of market institutions in the present model enforces the point made by Bryant, and also permits the inefficiency to manifest itself as involuntary unemployment.

The work on conjectural Keynesian equilibria (Benassy, 1977; Frank Hahn, 1978; Takashi Negishi, 1979) allows both flexibility of prices and binding quantity constraints, and obtains involuntary unemployment in models incorporating price determination. However, the basis for the conjectured demand curves assumed in this work is unclear. In particular, it is not obvious that quantities would or could actually respond to price changes in the manner that the price-setting agents conjecture they will. In contrast, equilibrium in the present model requires that the conjectures about the effects of changing one's actions correspond to the actual quantities that would result if such a change were made: perceived demands and supplies must be globally, rather than just locally, correct.

Finally, work based on efficiency wage models (for example, Steven Salop, 1979; Carl Shapiro and Joseph Stiglitz, 1984; Charles Kahn and Dilip Mookherjee, 1986) yields unemployment through job rationing that is an equilibrium response to informational asymmetries. For example, a positive level of unemployment arising through wages that exceed the supply price of the amount of labor actually employed may be necessary to provide incentives not to shirk. In such circumstances, equilibrium cannot involve full employment. Moreover, the unemployment in these models may not represent any inefficiency, given the informational constraints. More generally, efficiency wage

models seem best suited to explaining elements of the natural rate of unemployment. In the present model there are no problems of hidden knowledge or unobservable actions. Equilibrium may consequently be consistent both with full employment and with inefficient, "recessionary" unemployment.

The advantage of adopting the methodology used here is shown by the results that it permits. The costs are of two sorts. First, determining whether one has an equilibrium involves checking that no possible deviation is advantageous for any agent in any circumstance, and this is typically more time-consuming (although not mathematically more difficult) than verifying equilibrium conditions in a more standard model. Second, writing down an extensive form means that one has specified a very particular set of institutions. The ones assumed here do not seem to be a patently unrealistic representation of many actual markets, and they even match the informal descriptions in textbooks rather well. However, it is not immediately obvious that results proven for one set of institutions would hold under other reasonable specifications.⁵

In the next Section I describe the class of economies under consideration and the basic institutions and obtain the fundamental results. The succeeding section contains a discussion of extensions, alternative formulations, and other robustness issues, as well as suggestions for further work. The final section contains some open questions and tentative conclusions.

I. Unemployment with Perfectly Competitive Prices

Throughout we will consider a simple class of economic environments in which there are only five commodities and four types of agents, with $n \geq 2$ agents of each type.⁶ The

⁵In this regard, the work of Larry Jones and Rodolfo Manuelli (1987) on a variant of the model in Roberts (forthcoming 1988) is particularly pertinent.

⁶There is no real need to assume equal numbers of each type: what is important for the present analysis is that no agent is unique. (Roberts, forthcoming 1988,

commodities are called X , Y , R , S , and M . The first pair will be outputs, the second pair will be inputs, and M will not enter the production functions. The four types of agents are labeled A , B , J , and K , and superscripts will indicate a particular agent of a given type. The first two types of agents will be called employers, producers, or firms, although they will be treated as utility-maximizing agents. The second two are called consumers or workers.

A producer of type A (respectively, B) derives utility only from the consumption of M , of which he or she holds an endowment of μ_A (resp., μ_B). Only the producers have access to the technologies of production. This may be interpreted in terms of their alone being endowed with the relevant technical know-how or some other unmarketed factor. Type A agents have the technological knowledge to permit them to produce output X from input R , while the technology available to the type B agents allows production of output Y from input S . These technologies show constant returns to scale, and we set the input-output coefficients at unity. Neither type of firm is endowed with any of X , Y , R , or S .

These assumptions will mean that any profits received will be retained by the producers as M and that supplies and demands for the other goods will not be directly affected by the levels or distribution of profits. The lack of feedbacks from profits to demand simplifies the analysis considerably.⁷

Workers of type J (respectively, K) are each endowed with ρ of R and μ_J of M , (resp., σ of S and μ_K of M) and derive utility from Y , R , and M (resp., X , S , and M). Thus, J 's can supply input only to A 's and can buy output from only B 's. Correspondingly, K 's consume the output that A 's sell and supply the input used by B 's. In particular, no pair of agents has a mutually advantageous trade, each consumer is a

potential employee of only one type of firm and a customer of only the other, and each type of firm has only one type of consumer as potential employees and only the other as potential customers.

This separation of a firm's customers and workers is meant to model consumers' specializing in supplying labor but generalizing in consuming outputs. It has the effect of ensuring that a firm cannot directly increase the demand for its output by raising its employment or wages, because its customers' incomes are unaffected by such changes. Similarly, the supply of input (labor) to a firm is not directly dependent on its price and output levels. Some such separation or other source of leakage is important in obtaining involuntary unemployment equilibria. For further discussions of this setup, see Roberts (1987; forthcoming 1988).

We will assume that the preferences of each consumer of type J are represented by a utility function $U_J(y, \rho - r, m)$ while the consumers of type K have utility $U_K(x, \sigma - s, m)$. These functions are strictly quasiconcave, continuous, and strictly increasing for positive values, and take on values of $-\infty$ for $m < 0$.⁸ The utility functions for the firms are $U_A(m)$ and $U_B(m)$, which are also strictly increasing. A specific example is the case $\rho_J = \sigma_K = \mu_J = \mu_K = \alpha > 1$, $U_K = x + \ln(\alpha - s) + \ln(m)$, and $U_J = y + \ln(\alpha - r) + \ln(m)$. In this case, the unique Walrasian, perfectly competitive prices are $p_A = p_B = w_A = w_B = \alpha$, which result in each consumer buying $(\alpha - 1)$ units of output and selling $(\alpha - 1)$ units of input (see Roberts, 1987). Of course, in this case the firms earn zero profits and so consume only their initial endowments.

Given the economic environment, we must now specify the institutions for price determination, production, and exchange. In this section, we focus on a very simple set of such institutions, while in the next section

focuses on the case of $n = 1$.) The equal numbers assumption does simplify some of the arguments, however.

⁷These feedbacks are a major source of the multiplicity of equilibria in Heller (1986).

⁸Actually, it will be sufficient that utility is a very large negative number. The point is to simplify the arguments by ensuring that consumers will not risk going bankrupt. It will become clear later that a fear of bankruptcy is not what generates the unemployment equilibria.

we consider a variety of alternatives, extensions, and enrichments.

This base set of institutions begins with each firm stating a price for its output and a wage it will pay for its input.⁹ These are denominated in terms of M , the only universally desired good, which acts both as the unit of account and as the medium of exchange.¹⁰ Workers respond to these prices and wages by announcing the amounts of input they want to sell to each firm and of output they want to buy from each. Finally, the firms decide how much of these offers and orders to accept, production is carried out, and accounts are settled.

More formally, the institutions on which we focus involve each of the $2n$ firms announcing a single price for the good it sells and a wage for the input it purchases. These announcements are made simultaneously and independently. Then, knowing these choices, $p_A^1, \dots, p_A^n, w_A^1, \dots, w_A^n, p_B^1, \dots, p_B^n$, and $w_B^1, \dots, w_B^n$, each consumer J^i of type J announces a vector $y_J^{i1}, \dots, y_J^{in}$ of output amounts and a vector $r_J^{i1}, \dots, r_J^{in}$ of input amounts and, similarly, each K^i announces $x_K^{i1}, \dots, x_K^{in}$ and $s_K^{i1}, \dots, s_K^{in}$. Again the announcements are simultaneous and independent. Each of these quantities is to be interpreted as a bona fide offer to transact the amount of the good in question with the corresponding firm at the price or wage quoted by that firm. Thus, feasibility requires $\sum_j r_j^{ij} \leq \rho$ and $\sum_j s_j^{ij} \leq \sigma$ for all i : workers cannot offer to sell more than their available supplies.¹¹ Finally, knowing the announced prices and wages and the workers' directed offers and orders, each firm A^j selects $x_A^{1j}, \dots, x_A^{nj}$ and $r_A^{1j}, \dots, r_A^{nj}$ subject to

$x_A^{ij} \leq x_K^{ij}, r_A^{ij} \leq r_j^{ij}$ and $\sum_i x_A^{ij} \leq \sum_i r_A^{ij}$, and similarly for each type B firm. These quantities are the amounts of each order and offer that the firm accepts, and the constraints reflect conditions of voluntary exchange and feasibility. The final allocation then involves each consumer/worker J^i buying $\sum_j y_B^{ij}, i = 1, \dots, n$, selling $\sum_j r_A^{ij}$, receiving $\sum_j r_A^{ij} w_A^j$ in wage payments, and paying out $\sum_j y_B^{ij} p_B^j$ for output, and correspondingly for each type K consumer. Meanwhile, each A^i receives $\sum_j p_A^j x_A^{ji}$ in revenues and pays out $\sum_j w_A^j r_A^{ji}$ in factor payments, and similarly for each B^i .

A fuller discussion of this institutional specification is found in Roberts (1987; forthcoming 1988). A few major points are worth noting however. First, the determination of all prices and quantities as resulting from individuals' choices is explicitly modeled: there is no unmodeled market mechanism that somehow generates prices and quantities or brings about an actual allocation from individual quantity announcements. Second, any sequence of allowable action choices leads to a clearly defined outcome. These outcomes may involve rationing, but the rationing is a result of explicitly modeled choices as well. Third, when workers make their input supply and output demand announcements, prices and wages have already been chosen. This does not yet mean, however, that workers are price takers: this will be instead a property of equilibrium. Finally, prices and wages are flexible in that each firm may set the price and wage it controls at any level it wishes.

The game induced by the environment and institutions is assumed to be common knowledge among the agents, that is, there is no private information regarding tastes or endowments, no shocks or other exogenous uncertainty, and no confusion about the options open to the agents, the outcomes resulting from any specified list of actions, or the utility values assigned to outcomes.

A (pure) strategy for an agent is a complete contingent plan specifying the action to be taken by the agent in each possible circumstance.¹² These action choices can be

⁹ Thus we are assuming nondiscriminatory, linear pricing.

¹⁰ Thus M might be interpreted as commodity money. Of course, in this context there is no need to assume that M is "used up" in consumption. It is an open question whether a dynamic version of the model can be derived that would support the presence of M , interpreted as fiat money, in the utility function.

¹¹ The assumption that utility is arbitrarily negative for negative holdings of M means we need not worry about consumers' planning to buy more than they can afford.

¹² We do not consider mixed strategies.

contingent on the information available to the agent at the time the person takes the action in question. Thus, a strategy for firm A^i consists first of a particular specification of p_A^i and w_A^i , and then, *for each possible vector* of price and wage announcements and of input offers and output orders, a specification of r_A^{ji} and x_A^{ji} , $j=1, \dots, n$, meeting the feasibility and voluntary exchange constraints. Similarly, a particular strategy for a consumer of type J gives his or her choice of output orders and input offers as a function of the (announced) price-wage vector. Note that one player's strategy does *not* depend on any other's strategy, because strategies per se are not observed. Instead, a strategy makes actions depend only on observables, that is, on previously chosen actions that the agent in question has observed. Note too that a well-defined outcome results from *every* specification of the strategies for each agent and not just from those corresponding to some notion of equilibrium behavior. Thus, given conjectures as to the strategies being used by the others, each agent is able to evaluate the effect on his or her utility of selecting any particular strategy or deviating from it.

An equilibrium is formally a subgame perfect Nash equilibrium of the game induced by the environment and institutions. This requires that each agent correctly forecast the strategies the others are using, and that these strategies be best responses to each other at *every* decision point, *including those that would not be reached if behavior actually is generated by the specified strategies*.

This latter requirement of sequential rationality means that if one of the agents were to deviate from his or her prescribed strategy (for example, if one of the firms were to announce a different price and wage, or one of the consumers selected different amounts of input and output, than had been expected), the agents would still find it optimal to adhere from that point forward to the behavior specified by their equilibrium strategies. In particular, this means that the firms' quantity choices must maximize profit/utility for any price and wage vector and any quantity announcements from the workers. As well, at any price-wage vector

the workers must be behaving optimally, given the prices and wages and their (correct) conjectures about one another's choices and the firms' responses. This means that the workers in equilibrium must be price takers, treating the prices and wages as given and optimizing accordingly.¹³ In contrast, simple Nash equilibrium would allow workers to manipulate the price and wage choices by threatening to act in a manner that would force certain price-wage choices by the firms, even though, were these choices not made, adopting the threatened behavior would not be utility maximizing. Finally, of course, each firm's price and wage choice must be the utility/profit maximizing one for it, given its correct conjecture as to the other firms' choices and given its correct forecast that the consumers' responses to the different choices it might make will be given by their equilibrium strategies.

The major focus of this paper is on showing that particular price and wage vectors and allocations of goods can be the outcome of equilibria under the specified institutions. For example, in Proposition 1 we show this for the Walrasian prices, wages, and quantities. For an economy of the type considered here, a Walrasian solution¹⁴ is a price-wage vector (p_X, p_Y, w_R, w_S) and an allocation $\langle (x_A^i, r_A^i), (y_B^i, s_B^i), (y_J^i, r_J^i), (x_K^i, s_K^i), i = 1, \dots, n \rangle$ with the following properties:

- (1) (x_A^i, r_A^i) maximizes

$$U_A(\mu_A + p_X x - w_R r)$$
 subject to $x \leq r, i = 1, \dots, n$;
- (2) (y_B^i, s_B^i) maximizes

$$U_B(\mu_B + p_Y y - w_S s)$$
 subject to $y \leq s, i = 1, \dots, n$;

¹³Of course, they do not necessarily assume that they can buy or sell as much as they want at these prices. Rather, they recognize any quantity constraints that are implied by the other agents' strategies (see below).

¹⁴We avoid the more common terminology of "Walrasian equilibrium" to avoid confusion. A "Walrasian equilibrium" is not an equilibrium in our meaning, although (by Proposition 1) it is the outcome resulting from an equilibrium.

(3) (y_j^i, r_j^i) maximizes

$$U_j(y, p_j - r, \mu_j - p_Y y + w_R r),$$

$$i = 1, \dots, n;$$

(4) (x_K^i, s_K^i) maximizes

$$U_K(x, \sigma_K - s, \mu_K - p_X x + w_S s),$$

$$i = 1, \dots, n;$$

(5) $\sum x_A^i = \sum x_K^i, \quad \sum y_B^i = \sum y_J^i,$

$$\sum r_A^i = \sum r_J^i, \quad \sum s_B^i = \sum s_K^i.$$

Here (1) and (2) are utility (profit) maximization for the A and B types, (3) and (4) are utility maximization for the J 's and K 's, and (5) is market clearing.

PROPOSITION 1: *Given a Walrasian solution, there is an equilibrium under the institutions described above such that all firms announce the Walrasian prices and wages and the resulting allocation is the Walrasian one.*

Remark: To prove this result we will not actually specify directly the full equilibrium strategies at the outset. Instead, we will follow a different route which is more enlightening and simpler. It is based upon the observation that the natural way to solve the sort of game we are considering involves adapting the method used in dynamic programming to contexts with multiple-decision makers: start with the last decision points (here, the firms' quantity choices), determine equilibrium behavior at this stage in each possible circumstance, then roll back, solving for equilibrium at the next-to-last stage using the solution at the last stage to assign outcomes to each set of choices.

Specifically, what we do is specify the choices that are to be made "along the equilibrium path." We then must ensure that no agent wants to deviate unilaterally from these specified actions. This is a trivial matter at the last stage. However, at earlier stages, when the consumers are making their quantity choices and when the firms are picking prices and wages, evaluating the attractive-

ness of unilateral deviations requires that we specify the behavior that follows any such deviation. Of course, this behavior must be optimal for the agents involved. Thus, we have to specify the firms' responses if any single consumer, when faced with the equilibrium prices and wages, makes quantity announcements differing from the specified ones. As well, we must also specify the quantity responses from consumers and the hiring and output decisions of the firms that follow any single firm's deviating from the specified price and wage announcement.

To complete the analysis, in situations involving multiple deviations we can make any specification of behavior so long as the strategies are optimal against one another from each point forward. In this regard, it is useful to note that for any vector of price and wage announcements from the firms, one equilibrium mode of behavior is for each consumer to announce zero and zero supply. With all other consumers ordering and offering zero, no firm to which a consumer might offer labor is receiving any orders, and so it would not hire, while no firm from which the consumer could order is offered any labor, and so it will not be able to produce to meet the expressed demand.

PROOF:

Let (p_X, p_Y, w_R, w_S) , $\langle (x_A^i, r_A^i), (y_B^i, s_B^i), (y_J^i, r_J^i), (x_K^i, s_K^i), i = 1, \dots, n \rangle$ be the Walrasian solution. Note that $p_X \leq w_R$, with equality if $x_A^i > 0$ for some i , and $p_Y \leq w_S$, with equality if $y_B^i > 0$ for some i . Also, $x_A^i = r_A^i$ and $y_B^i = s_B^i, i = 1, \dots, n$. Select $4n^2$ nonnegative numbers $x^{ij}, y^{ij}, r^{ij}, s^{ij}, i = 1, \dots, n, j = 1, \dots, n$, so that $\sum_j x^{ij} = x_K^i, \sum_i x^{ij} = x_A^j, \sum_j y^{ij} = y_J^i, \sum_i y^{ij} = y_B^j, \sum_j r^{ij} = r_J^i, \sum_i r^{ij} = r_A^j, \sum_j s^{ij} = s_K^i$, and $\sum_i s^{ij} = s_B^j$.

Assume each type A firm has announced $(p_A^i, w_A^i) = (p_X, w_R)$, each B has announced $(p_B^i, w_B^i) = (p_Y, w_S)$, each J^i has made offers $(r^{i1}, \dots, r^{in})$ and orders $(y^{i1}, \dots, y^{in})$, and each K^i has announced $(x^{i1}, \dots, x^{in})$ and $(s^{i1}, \dots, s^{in})$. It is clearly equilibrium behavior for the firms to hire the amounts offered and meet the demands they receive, because all feasible quantity choices for the firms yield them nonpositive profits and these particular ones yield zero. If, given the Walra-

sian prices and wages, consumer choices differ from those specified, let the firms simply maximize given the actual orders and offers. Then no consumer will gain by unilaterally deviating from the specified quantity choices because these are utility maximizing. Thus, we have equilibrium behavior given the Walrasian prices and wages.

Since the Walrasian price and wage are equal, no firm can gain by cutting its price while raising its wage. Then to ensure that no firm wishes to charge different prices and wages, it is sufficient to specify that if any firm deviates from the Walrasian price and wage except by raising its wage and lowering its price, then all offers to and orders from it are zero. In such circumstances, the workers and consumers who deal with the deviating firm when it announces the Walrasian price and wage now deal with other firms and have their Walrasian orders and offers filled.

This consumer behavior is clearly consistent with equilibrium if the deviating firm has raised its price and lowered its wage. However, if the firm has either raised both its price and wage or lowered both, matters are more complicated. If both are increased, all the workers of the relevant type would like to deal with the deviating firm but customers would prefer to buy elsewhere at the lower Walrasian price. If both the price and wage are lowered, consumers would benefit from dealing with the deviator but workers would rather find employment elsewhere at the Walrasian wage. Given the wage-price deviation, equilibrium in the continuation game clearly requires that workers and consumers effectively coordinate, with workers offering labor to the deviating firm if and only if consumers place output orders with it. However, given the deviation, either pattern can represent optimal behavior: If either group (workers or consumers) expects the other to place orders or offers only with the deviator, optimal behavior requires dealing with that firm, too, while if either group is expected not to deal with the deviator, the other should not place offers or orders with it either. The situation thus has elements of the Battle of the Sexes. To ensure that the firm has no incentive to deviate from the Walrasian price and wage, we resolve the conflict as above, that is, in favor

of the workers if the price and wage are lowered, and in favor of the consumers if they are increased.

Finally, to obtain the Walrasian solution as an equilibrium outcome, we simply specify any subgame equilibrium at price-wage vectors at which more than one firm is deviating from the Walrasian levels. Note that such subgame equilibria exist; in particular, all offers and orders being zero is always one such. This follows because, with zero orders for output, no consumer can ever sell any labor and so might as well offer zero, and with zero offers, a positive output order can never be filled.

Note that although the outcome of this equilibrium is perfectly competitive, the behavior is not. Instead, each agent recognizes the quantity constraints implied by the others' choices. Of course, under the given equilibrium, these constraints are not binding. However, other equilibria exist in which they do bind, and this is the key to obtaining equilibrium unemployment at competitive prices.

PROPOSITION 2: *Given a Walrasian solution, let $0 \leq k \leq n$. Then there exists an equilibrium under the specified institutions in which all firms announce the Walrasian prices and wages and in which the equilibrium outcome gives k of the consumers of each type their Walrasian allocations and the $(n - k)$ remaining consumers of each type receive their initial endowments, buying and selling zero amounts.*

PROOF:

Let $(p_X, p_Y, w_R, w_S), \langle (x_A^i, r_A^i), (y_B^i, s_B^i), (y_J^i, r_J^i), (x_K^i, s_K^i), i = 1, \dots, n \rangle$ be the Walrasian solution. Note that, with strictly convex preferences, all consumers of a given type receive the same consumption bundle at the Walrasian solution, that is, $(x_K^i, s_K^i) = (x_K, s_K)$ and $(y_J^i, r_J^i) = (y_J, r_J)$ for all i . As well, all producers end up with their initial endowments.

Note that there is a Walrasian outcome in which all the firms of each type produce identical amounts. In this case, we can think of each A^i as buying only from J^i and selling only to K^i and, correspondingly, each

B^i as dealing only with K^i and J^i . We will prove the result for this case; the argument for the general case will then be clear. The argument follows the same line as that in Proposition 1.

Suppose all firms have announced the Walrasian prices and wages. Let the k agents of each type who are to be active be indexed by $1, \dots, k$. For each $i \leq k$, let J^i offer to sell r_j of R to A^i and order y_j of Y from B^i , let K^i offer s_K of S to B^i and order x_K of X from A^i , and let this pair of agents' orders from and offers to firms $j \neq i$ be zero. For $i > k$, let J^i offer zero of R and order zero of Y from each firm and, similarly, let K^i 's orders and offers all be zero. Note that these transactions are feasible. Further, it is optimal for each firm to hire the labor offered to it and fill the orders it receives.

To show that this pattern of offers and orders, with the firms' hiring all the labor offered and meeting demand, is equilibrium behavior in the subgame given the Walrasian prices and wages, we must show that no consumer can gain by unilaterally deviating from the specified orders and offers, given the wages and prices and given that the firms respond optimally in the face of such behavior. That this holds for $i \leq k$ is obvious: each such J^i and K^i is realizing his or her utility-maximizing trade. For $i > k$, a firm to which worker/consumer i offers labor sees no demand for the extra output that hiring i would yield and, if active, sees no gain to hiring i in preference to its assigned worker. Similarly, a firm from which i orders does not have the input available to meet this extra demand and, if active, gains nothing from "deserting" its regular customer for i . Thus, we can specify that hiring and production remain unchanged from the pattern specified above if any J^i or K^i , $i > k$, deviates from zero orders and offers. Thus, no consumer gains by deviating.

Now we must ensure that no firm wishes to deviate from the Walrasian price and wage. The argument is essentially the same as in Proposition 1.

Again, note that wage increases coupled with a price cut cannot be profitable for any firm, so that we need consider only price increases coupled with wage decreases and movements of both the price and wage in the

same direction. In the former case, suppose that A^i announces $p_A^i \geq p_X$, $w_A^i \leq w_R$, with at least one inequality being strict. Then we specify that the offers to A^i and the orders placed with it are zero. If $i > k$, so that A^i is inactive, this can be accomplished as under Walrasian prices, because zero orders from and offers to any firm are optimal for each consumer/worker if all others are ordering and offering zero. In this case, the price and wage change do not affect the firm's activity level. If $i \leq k$, so that A^i is active at $p_A^i = p_X$, $w_A^i = w_R$, then we specify that J^i shifts his or her offer of the Walrasian quantity of R and K^i shifts his or her demand for X to some firm A^j , $j \neq i$, while all other orders and offers are as at the Walrasian price and wage. The deviator A^i thus loses all its business. Further, A^j hires J^i and meets the demand from K^i , as is optimal for it. Thus, no such deviation in price and wage is profitable.

If some firm, say A^i , either lowers both its price and wage or raises both, then the conflict noted in the proof of Proposition 2 between the K 's as consumers and the J 's as workers arises again. In either case, we again specify that the orders and offers placed with A^i be zero. If $i > k$, this is again accomplished by having the pattern of orders and offers be as at the Walrasian price and wage. This constitutes an equilibrium in the subgame, again because zero offers to and orders from a given firm always constitute best responses to one another. Either all the J 's would like to deal with A^i (if $p_A^i \geq w_A^i > w_R$), or all the K 's would like to do so (if $w_A^i \leq p_A^i < p_X$), but in each case, if the one group expects the other to shun A^i 's non-competitive price and wage, so will it. Meanwhile, if $i \leq k$, again J^i and K^i coordinate in dealing with some A^j , $j \neq i$.

Finally, specifying any subgame equilibrium outcome at price-wage vectors where two firms are deviating from the Walrasian levels, we have an equilibrium with unemployment.

The key step in this argument (and in that establishing Proposition 1) is that a firm that deviates from the competitive price and wage ends up with zero activity. This was accomplished, in effect, by having the supply of

labor to any firm be infinitely elastic with respect to wage cuts below the level offered by other potential employers and by having the output demands be infinitely elastic with respect to price increases. This seems to be a natural general equilibrium analogue of Bertrand competition in both markets. Of course, other forms of behavior are conceivable, and some of these might well eliminate the involuntary unemployment of Proposition 2. What is unclear is whether any of these would support a full employment equilibrium and the Walrasian outcome in particular.

II. Alternatives and Extensions: Robustness of Results

The results in Section I indicate that a natural specification of equilibrium behavior may lead to markets' failing to clear in the context of institutions for price and quantity determination that do not seem to be an unreasonable formalization of how one might think these processes actually occur. Nevertheless, if there were natural modifications either of behavior or of the institutions that would ensure full employment, or if the failure of market clearing were crucially dependent on particular, unsatisfactory assumptions on the structure of the economy, the lack of robustness of the results would limit their interest. Thus, in this section we begin an exploration of the robustness issue.

A natural first response to the unemployment result is to ask why unemployed workers do not approach active firms and underbid the workers they are hiring.¹⁵ However, allowing workers such an option does not obviously resolve the problem of unemployment because one would expect that the workers who are threatened with losing their jobs would respond by further undercutting. To see if this does actually eliminate unemployment, we must formalize

these possibilities by specifying explicitly institutions that include them.

One natural place to insert the possibility of workers' offering to work for less is after the firms have made their quantity decisions. Suppose then we allow each worker at this stage to approach some firm and offer to work at a lower wage than the firm had earlier quoted, but that, if a firm is approached in this fashion, the worker(s) it was going to hire can make a counteroffer. The firm then will have the option of accepting either, both, or neither of these offers. Our object is to show that the unemployment equilibria described in Proposition 2 survive this alteration in the institutions.

To complete the institutional specification, the process for determination of quantities after such underbidding must be given. There are a wide variety of options here, both with regard to the amount of labor to be provided when undercutting occurs and the amounts of output to be supplied.

On the input side, the simplest solution is to require that, if underbidding occurs, the workers doing so must offer to supply the same amount of labor that the firm had originally planned to hire, even though this amount might exceed the workers' competitive supply at the lower wage. Given the orders it faces, the firm will then hire the same amount as before. Note that if the workers offered some different quantity, the firm might change its output level, and the rational forecast of this would alter offers and orders throughout the economy.

Given this requirement on input supplies, the simplest rule on the output side is that workers do not get to revise their quantity orders from those made earlier. Note, however, that if underbidding does occur in equilibrium, then this will be forecast and the orders placed will allow for the reduced wages and the hiring that results.

In this context, it is easy to see that the possibility of undercutting may have no effect. In particular, consider the case in which the workers' endowments are $\mu_J = \mu_K = \rho_J = \sigma_K = \alpha$ and the utility functions are $U_J(z, t, m) = U_K(z, t, m) = z + \ln(\alpha - t) + \ln m$. In this case, the Walrasian quantities are $(\alpha - 1)$ and the price and wage levels are

¹⁵ Recall that we have already dealt with firms' cutting wages, although we have not allowed discriminatory wage offers. However, it seems clear that even discrimination would not upset the unemployment outcome.

α . Then, if an agent of type J , say, has initially ordered $y = 0$ and has sold no labor, the lowest wage he or she would accept if he or she must provide $(\alpha - 1)$ units of labor and still get no output is $\underline{w} = \alpha$, that is, the prevailing Walrasian wage. Thus, such an agent in fact will not underbid the Walrasian wage, and the equilibrium described earlier survives, even with no counteroffers from the employed.¹⁶

Of course, this specification may not seem a fair representation of the idea of undercutting, because the only reward to working is in the increase in M and not in increased consumption of output. At the opposite extreme, we could allow all agents to revise their quantity decisions after any underbidding, but this would lead to a highly complicated model. A simpler alternative is that if one worker ends up supplanting another through underbidding, then the former simply takes over (and pays for) the output order the other had previously placed.

In this case, suppose the strategies for the initial stages are as in Proposition 2, so that the prices and wages quoted by each firm are equal, the quantities bought and sold by agents $1, \dots, k$ are the Walrasian amounts and the quantities transacted by the other agents are zero. Clearly no employed agent can gain by underbidding. Consider then an unemployed agent, say J^i , $i > k$. Since no firm A^j , $j > k$, sees demand for its product, it is pointless to offer to work for such a firm. Let J^i then approach some active firm, A^j , $j < k$, and offer to work for w^i , which is less than the Walrasian wage, w^W . Letting r^W and y^W denote the Walrasian quantities of R and Y and p^W the Walrasian price of Y , the lowest value that i would be willing to set for w^i is given by $U_j(y^W, p - r^W, \mu_j - p^W y^W + w r^W) = U_j(0, p, \mu_j)$. This same value also defines the lowest counteroffer that J^j , the worker tentatively assigned to A^j , would make.

It is easy to see that equilibrium requires that A^j accept the lower of the wages offered by J^i and J^j . Then, the logic of Bertrand competition means that the unique equilibrium in the subgame is for both J^i and J^j to bid $\underline{w}$. But then, even if hired, J^i gains nothing over being unemployed, and so we may specify that the person never bothers to underbid in the first place. Thus, the allocation from Proposition 2 remains an equilibrium outcome, even when underbidding is permitted.¹⁷

Another approach to allowing workers to influence wages is simply to have the firms at the first stage announce only output prices, while each worker announces the wage at which the person is willing to sell his or her labor. It would then be natural to have each firm announce an amount of labor it wants to hire from each worker and, simultaneously, each consumer announce an amount of output he or she wants to buy from each firm. Finally, workers could decide how much of each employment offer to accept, and firms could then decide on how much to produce and sell.

Note that a firm anticipating zero orders will make no offers of employment, and if consumers expect a firm not to hire there is no sense placing orders with it. Thus the arguments used earlier can be adapted to obtain a result paralleling Proposition 2 for these institutions.

A second possible alteration in the setup involves agents' behavior within the original institutions. The unemployment equilibria in Proposition 2 involve unemployed workers' ordering zero quantities of output and offering zero quantities of labor. This is rational (that is, equilibrium) behavior; no individual can unilaterally change orders and offers and be better off as a result. Still, these orders and offers may well be less than the workers' "effective demands," defined as maximum

¹⁶This argument will clearly work so long as $U(0, \tau - t, \mu + wt) \leq U(0, \tau, \mu)$, where τ and μ are the endowments and t and w are the Walrasian labor supply and wage, that is, as long as $U_2/U_3 > w$ at $(0, \tau - t, \mu + wt)$.

¹⁷This argument obviously pushes the Bertrand logic very hard. Nevertheless, it does not seem totally improbable to suppose that an unemployed worker might not expect to be able to supplant an employed one by offering to work for less because the latter will be willing to match his or her offer.

amounts that the individual would be willing to transact on each market, given that the person forecasts that the transactions on the other market will be constrained to zero (Benassy, 1973).

Benassy (1982) has argued that, in such a situation, agents should announce their effective demands, rather than their anticipated transactions, because the latter too easily lead to unemployment equilibria. Realizing beneficial trades requires that agents somehow transmit the information that they are willing to trade, and announcing anticipated transactions may fail to do this. Moreover, in the present context it would seem costless to increase one's orders and offers, given the behavior of the others, while if others also did so, the unemployment might be broken.

While Benassy's effective demands have been questioned on various grounds (Ito, 1979; Gale, 1983; Roberts, forthcoming 1988), we wish to show that, even if agents do announce effective demands, equilibrium may still fail to yield market clearing.

Consider again the particular utility functions and endowments discussed above and recall that the Walrasian prices and wages are all equal to α . Now consider the effective demands of a worker at these prices and wages who anticipates making zero transactions. For output, these are given by the solution of

$$\max_z z + \ln \alpha + \ln(\alpha - \alpha z),$$

which is uniquely $z = 0$, while the effective supply of input is the solution of

$$\max_z \ln(\alpha - z) + \ln(\alpha + \alpha z),$$

which is $(\alpha - 1)/2$.

Thus, in this case, if workers announce their effective demands for input and output, they actually do indicate that they are willing to work at the going wage. However, demands are still zero, so hiring is zero. From this, we see that unemployment of the type demonstrated in Proposition 2 still may

persist even if workers state their effective demands.¹⁸

It may be worth noting that, if workers were all to announce their Walrasian supplies and demands, then full employment would be achieved (this is just Proposition 1), but this behavior is, in a sense, risky. In the context of the example, suppose one individual announces his or her Walrasian demand and supply but the "corresponding" worker-consumer of the other type announces a positive labor supply and a zero output demand. Then the first agent can end up with negative money holdings, because the person is committed to buying output but sells no labor. This argument suggests that out unemployment equilibria would not be eliminated by requiring "trembling hand" perfect equilibria (Selten, 1975) rather than just subgame perfect ones.

While the preceding results formalize the "Keynesian" intuition that expectations of low levels of demand can be self-confirming and can result in equilibria with involuntary unemployment, they do not exactly match with earlier discussions of these ideas (Clower, 1965; Leijonhufvud, 1968). In that literature, it was the expectations held by each firm about the demand that would be forthcoming at given prices as it varied its hiring that were central; here firms already have received output orders when making their hiring decisions and so need make no such forecasts. Instead, workers are attempting to predict one another's quantity choices, and the inefficient equilibria arise from their pessimism.

However, a modification of the assumed institutions yields a more central role for firms' beliefs and captures much of the standard view while still maintaining the possibility of unemployment equilibria.

¹⁸Again it is clear that this argument would work more generally and, in particular, if either the effective demand for output when constrained to sell no labor or the effective supply of labor when constrained to buy no output is zero. For this it is sufficient that either $U_2/U_3 > w$ or $U_1/U_3 < p$ at (o, τ, μ) .

Specifically, suppose that, as before, each firm first announces a price and wage. Next, however, workers make (directed) announcements not of both their labor supply and output demand but, instead, just of amounts of labor they wish to sell. Then, knowing the workers' labor offers, each firm independently makes its hiring decisions, which consist, in each case, of how much of the offers made to it to accept. At this point, the firms become committed to paying the contracted amounts.¹⁹ Next, the workers place output orders, knowing the amount of labor they have sold and the wages they will receive. Finally, firms decide how much of these to fill. Assume, as before, that at each stage all agents are fully informed about all previously made choices.

An equilibrium under these institutions is again a subgame perfect equilibrium in the induced game, and again we find such equilibria by examining the stages sequentially, beginning with the last.

The determination of output levels at the last stage is completely trivial, because the firms' costs are, at this stage, fixed. Thus, we can assume that each will always produce the smaller of the total ordered from it and its maximum feasible output (as determined by its hiring). Note, however, that if production is smaller than orders, the firm will have to decide how output is to be allocated among those ordering from it. Of course, the firm is indifferent over all such distributions of the given output, so any pattern of such rationing is consistent with equilibrium in these subgames.²⁰

In equilibrium, consumers will correctly foresee the allocation of output resulting from any specified list of output orders.

¹⁹ Feasibility may thus require that the firms' hiring be constrained by their endowments of M .

²⁰ Note that if orders are smaller than hiring, the firm need not use all the labor for which it has contracted to pay. In the sequel we will assume, for simplicity, that if a firm has hired excess labor, the full amount must be supplied by workers and is unavailable for their own consumption. Of course, equilibrium will involve hiring equal to sales.

Using this forecast and knowing prices, wages, and hiring, each consumer announces an output order to maximize his or her utility, given conjectures about the offers others are placing.²¹ Equilibrium then requires that these conjectures are correct, so that each consumer's order is optimal against the orders that others are actually placing.

A particular specification of equilibrium at this stage then allows each firm to decide how much to hire by examining the sales that result from various hiring levels, given its conjectures about the other firms' hiring. Again, subgame equilibrium is characterized by each firm's choice being optimal, given correct conjectures about the others' decisions and correct forecasts regarding the resultant orders and sales. Once more rolling back, workers then can forecast the full implications of making different patterns of labor supply offers. Solving for a subgame equilibrium here then gives the relations between prices, wages, and quantities that firms use in deciding on their price and wage announcements.

The sequential determination of inputs and outputs has important consequences here. Not only are workers sure of their incomes when they place their output orders, but also they know the supplies potentially available from the various firms. This can, in some circumstances, break the zero-activity level equilibria that involve each consumer ordering zero because the person expects no labor to be supplied and thus no output to be available, and simultaneously offering no labor because he or she expects zero output demand and hiring. Suppose now a firm has announced a price at which demand is positive even when its customers cannot sell any labor. The firm can then anticipate that if it hires it will receive demand. Consequently, workers know that if the firm's wage is lower than this price, it will in fact do some hiring if it is offered labor. They will then make positive labor supply offers so long as doing

²¹ Existence of an optimal order may depend on the rationing rule: see Benassy (1982).

so is optimal given the firm's wage, the prices they face for output, and their forecasts of output availability.

Nevertheless, equilibria parallel to those in Proposition 2 may still exist with these institutions.

PROPOSITION 3: *Consider the economy with endowments $\mu_J = \mu_K = \rho_J = \sigma_K = \alpha$, $\alpha > 1$, and consumer preferences $U_J(z, \alpha - t, m) = U_K(z, \alpha - t, m) = z + \ln(\alpha - t) + \ln m$. Let the institutions be specified as above with sequential supply offers and output orders. Then for each k , $0 \leq k \leq n$, there exists an equilibrium in which all firms announce the Walrasian price and wage levels, consumers $1, \dots, k$ of each type transact their Walrasian quantities and consumers $k + 1, \dots, n$ transact zero amounts and consume their initial endowments.*

PROOF:

As in Proposition 2, let the agents of each type who are to be active be indexed as $1, \dots, k$, while the unemployed are indexed by $k + 1, \dots, n$.

We again limit ourselves to describing the actions chosen along the equilibrium path and to ensuring that there are never incentives to deviate. These actions are: all firms announce prices and wages equal to the Walrasian level, α ; consumer i of type J (respectively, K) offers $(\alpha - 1)$ of R to A^i (resp., of S to B^i) if $i \leq k$ and offers an amount less than or equal to $(\alpha - 1)/2$ of R to A^i (resp., of S to B^i) if $i > k$; each firm i hires the full amount offered to it if $i \leq k$ and hires 0 otherwise; consumer J^i orders $(\alpha - 1)$ of Y from B^i (resp., K^i orders $(\alpha - 1)$ of X from A^i) and orders zero from all other firms if $i \leq k$, while, for $i > k$, J^i and K^i order zero from all firms; and, finally, each firm fills its orders.

Clearly, the firms' behavior at the last stage is optimal, as is that of the consumers $1, \dots, k$ of each type at the order stage. As well, because (as noted earlier) the optimal purchase of output at a price of α is zero if one has sold no input, the consumers with $i > k$ are also acting optimally in their orders.

The hiring of the firms yields each of them zero profits, and no deviation from this be-

havior can generate positive profits with prices and wages set at the Walrasian levels. Thus, the specified hiring behavior is consistent with equilibrium.

Obviously, given the prices and wages, no active consumer can benefit from changing his or her labor offer. If an inactive worker changes either the size of the offer or the firm to which it is made, then we specify that the person still sells zero. That this is consistent with equilibrium is simply seen by noting that if the firm hires any positive amount from this worker, then it earns zero if it manages to sell the resultant output (and so does not gain) and otherwise loses money.

Thus we arrive at the stage where prices and wages are set.

Consider first an active firm that considers deviating from the Walrasian price and wage. If this deviation involves a wage cut, then we simply specify that the firm receive no labor offers. (Any worker who would have offered labor to this firm now goes to some other, nondeviating firm and offers it his or her Walrasian supply. This is accepted, and any customer who would have bought from the deviator along the equilibrium path now deals with the firm with the increased hiring.) Thus, no wage cut helps.

Suppose then the deviator, say firm A^i , raises its wage. Of necessity, it will have to raise its price even more to gain from this deviation. Let its choices then be $p > w > \alpha$. We must now ensure that this gets zero sales. Note that we cannot simply adopt the technique used in Propositions 1 and 2 of having the workers expect the demand all to go to firms charging the low, Walrasian price, because here workers move first. Thus, if all of them go to the high-wage deviator, it alone will have output and customers will have no effective option but to deal with this firm. Nevertheless, it still turns out to be an equilibrium for workers to shun the high-wage deviator, in essence because they expect it will not, in fact, hire them because it correctly forecasts a zero demand for its product.

The specified response to an A setting $p > w > \alpha$ is then for all workers to make Walrasian offers to nondeviating firms, for these to be accepted, for consumers each to

order their Walrasian quantities, placing the orders in such a way that they can be met, and for the nondeviators to fill the orders. In particular, if the deviator is A^i , let J^i and K^i , $i \neq j$, deal with A^i and B^i , while J^j and K^j deal with A^{j+1} and B^{j+1} if $j < n$ and with A^1 and B^1 otherwise.²²

The key issue is whether any worker-consumer can gain by deviating from this pattern. Clearly none can gain by changing his or her order, given the specified pattern of labor offers and hires, nor can any K gain from changing his or her offer of S . But can a worker of type J offer labor to the deviating, high price and wage A and get more than the Walrasian utility level?

Suppose first that the deviator is an active firm, say A^1 , and that J^1 offers labor to A^1 rather than A^2 . (The case in which J^j , $j \neq 1$, offers labor to A^1 is identical.) The other workers must continue the offers specified above because their offers are made simultaneously with J^1 's, but the firms will be able to adjust their hiring in response to this deviation and the consumers will be able to alter their orders.

Suppose then that A^1 hires r^1 from J^1 . Let A^i continue to hire $(\alpha - 1)$ from J^i , $i \geq 2$, let B^2 continue to hire $(\alpha - 1)$ from K^2 and zero from K^1 , and let B^i hire $(\alpha - 1)$ from K^i , $i > 2$. (Recall that, following the deviation by A^1 but absent the subsequent deviation by J^1 , K^1 would be employed by B^2 . Also recall that B^1 receives no offer of labor when A^1 deviates.) Now, let consumers J^i (resp., K^i) order their optimizing quantities from B^i (resp., A^i), $i \geq 2$, let K^1 order his optimal quantity, which, because he or she is not hired, is zero, and suppose that if J^1 places an order with any of the firms $B^1, \dots, B^n$, it is not filled. Note that this behavior for the firms is optimal because B^1 has no output to offer and each B^i , $i \geq 2$, is selling its entire output and is indifferent about who receives it. Then the best that J^1 can do is receive 0 output. Further, since K^1

is not hired, firm A^1 will get no demand and so will actually not accept J^1 's offer. Thus, J^1 ends up with his or her initial endowment and is worse off for deviating. This in turn means no active firm has an incentive to deviate from the Walrasian price and wage.

Finally, a similar argument shows that A^i , $i > k$, does not gain by raising its price and wage because, again, no worker-consumer will optimally deal with it in preference to making his or her Walrasian transaction with another firm.

Thus we have the possibility of some fraction of the economy enjoying its Walrasian consumption vectors while the remainder of the agents end up in equilibrium unable to transact at all, even though these unemployed workers explicitly state their willingness to work at currently offered wages and are in all respects identical to the employed. Moreover, the reason they are not hired is that firms simply do not forecast sufficient demand. At the same time, there is also an equilibrium at the same Walrasian prices and wages with full employment. However, the decentralized market institutions do not effectively coordinate individual actions and ensure that the efficient equilibrium is reached.

There are many obvious directions for further extensions of this work. We mention only four of these.

First, a central feature of the proof of the existence of unemployment equilibrium in Proposition 3 is that full employment obtains if one of the firms raises its price and wage above the Walrasian level. (Note that this is not the case in Proposition 2, nor is this pattern employed in Proposition 3 if the firm makes the possibly more intuitive deviation of cutting its wage offer in the presence of unemployment. In these cases, a deviation simply results in the firm's losing business, as under Bertrand competition, but not in a change in the aggregate level of activity.) Of course, the firm itself does not gain from this sort of deviation, so unemployment remains as an equilibrium phenomenon. However, it would clearly be interesting to see if unemployment can be sustained under sequenced orders and offers without there being higher

²² Note that we thus have full employment "off the equilibrium path" if a firm deviates in its price and wage choice by raising its wage. We will return to this below.

levels of employment off the equilibrium path than along it.

More generally, it can be argued that the responses to a wage increase assumed in Proposition 3 involve too great an economy-wide reaction to a single firm's actions: recognition of observability questions and the need for search in real economies argue for some measure of continuity. There is certainly merit to these points, and it would be very interesting to investigate whether continuity can be imposed while maintaining the existence of both full employment and unemployment equilibria. Results of this sort are given an example of a world of monopolies (that is, $n = 1$) in Roberts (forthcoming 1988), but none are yet available for the competitive environments considered here.

A further feature of this proof (which also was important in Propositions 1 and 2) was the heavy use made of the firms' technologies' showing constant returns to scale. This implied both that the firms earned zero profits in competitive equilibrium and that they were willing and able to expand to absorb the workers and customers of a competitor at the given price and wage. If nonconstant returns are assumed, firms will no longer be indifferent regarding output levels at the Walrasian prices and wages. In this case, many aspects of the arguments made above would fail. Clearly, it would be interesting to determine whether equilibrium under decreasing returns is still consistent with both full employment and unemployment at Walrasian prices and to study the nature of equilibrium with increasing returns.

In a related vein, one should relax the special structure of endowments and preferences assumed here. It seems that some separation of a firm's potential employees and potential customers is important for coordination failures, but how much such separation is needed? A more general model might allow workers to be interested in a wide variety of outputs, so that their demand would be a small fraction of the total demand facing their employer. It seems likely that results paralleling those obtained here would then hold for "approximate equilibria" in which each firm ignores the impact of its hiring and wage payments on its demand, but it is unclear if this would be the

case for full equilibria. As well, it would be worthwhile to allow workers to be able to supply labor to several different industries. As long as they continued to be uninterested in consuming any output to whose production they could contribute, this should lead to no alteration in the results.

The fourth desirable extension would be to incorporate the possibility of government which could try to use planning or fiscal policies to lead the economy to efficient equilibria. More generally, one would like to investigate whether there are other alterations in the institutional framework that seems natural and would yield more efficient outcomes.

III. Conclusions

The question of whether involuntary unemployment can be an equilibrium phenomenon has troubled economists for at least 50 years. If we use standard models in the tradition of Cournot and Walras, the answer must be that equilibrium implies market clearing. But the process by which prices and quantities are determined is not fully specified in such models, and, in effect, market clearing is made part of the definition of equilibrium.

In this paper I have attempted to be fully explicit about these processes and then have shown that the natural concept of equilibrium under reasonable institutions need not imply market clearing: absent a Walrasian auctioneer or some similar institution, the decentralized decisions of individuals may be left inefficiently uncoordinated by market forces, even though all agents are very well informed, have fully rational expectations, and maximize accordingly, and even though prices and wages are flexible.

The analysis presented here points to a crucial role for expectations or "animal spirits." This role is very much in accord with Clower's treatments of Keynesian ideas and with later developments of his ideas by others concerned with macroeconomic issues. Thus, I have focused the discussion of equilibrium without market clearing on involuntary unemployment. There is still, however, a serious question of whether the present analysis has any real power in helping us

understand actual unemployment, business cycle activity, and the like. The problem is that the model generates too many equilibria: we have the range of equilibria at Walrasian prices identified in Propositions 2 and 3, and it is possible that there might be others as well.²³ While a multiplicity of equilibria seems central to any formalization of Keynesian ideas, the extreme multiplicity present here deprives the model of predictive value.²⁴

Nevertheless, the ease with which we have obtained equilibrium without market clearing, once the institutions for price and quantity determination are explicitly modeled, seems to me to raise important questions. Is it simply that the institutional specification given here is the wrong one, and that market clearing would result automatically if the right institutions were assumed? A centralized Walrasian market would have this property, but assuming these institutions actually exist seems ridiculous. Or, perhaps, is there some equilibrium selection mechanism that leads markets to the most efficient equilibria but that has been ignored here? The fixed-price literature in effect seems to assume something like this in positing that, given prices, the rationing is efficient, but the forces that would bring this about are not obvious. Or, perhaps, have we simply been too hasty in assuming that our models with built-in market clearing will accurately predict actual market outcomes?

²³In particular, it is an open question whether there can be equilibria in which firms announcing prices and wages different from the Walrasian ones are active. The logic of Bertrand equilibrium might suggest that there ought not to be such equilibria, but the issue seems complex.

²⁴Note that there is still the possibility that the presence of autonomous government demand might yield more clear-cut results.

REFERENCES

- Barro, Robert J. and Grossman, Herschel I., "A General Disequilibrium Model of Income and Employment," *American Economic Review*, March 1971, 61, 82–93.
- Benassy, Jean-Pascal, *Disequilibrium Theory*, doctoral dissertation, University of California, Berkeley, 1973.
- , "A Neo-Keynesian Model of Price and Quantity Determination in Equilibrium," in *Equilibrium and Disequilibrium in Economic Theory*, G. Schwodiauer, ed., Dordrecht: D. Reidel Publishing, 1977, 511–44.
- , *The Economics of Market Disequilibrium*, New York: Academic Press, 1982.
- Bryant, John, "A Simple Rational Expectations Keynes-Type Model," *Quarterly Journal of Economics*, August 1983, 98, 525–28.
- Clower, Robert W., "The Keynesian Counter-revolution: A Theoretical Appraisal," in *The Theory of Interest Rates*, Frank H. Hahn and Frank P. R. Brechling, eds., London: Macmillan, 1965.
- Cooper, Russell and John, Andrew, "Coordinating Coordination Failures in Keynesian Models," Cowles Foundation Discussion Paper 745, Yale University, 1985.
- Diamond, Peter, "Aggregate Demand Management in Search Equilibrium," *Journal of Political Economy*, October 1982, 90, 881–94.
- Drazen, Alan, "Involuntary Unemployment and Aggregate Demand in an Optimal Search Model," Working Paper No. 32-86, Foerder Institute for Economic Research, University of Tel Aviv, 1986.
- Drèze, Jacques, "Existence of an Exchange Equilibrium Under Price Rigidities," *International Economic Review*, June 1975, 16, 301–20.
- Gale, Douglas, *Money: In Disequilibrium*, New York: Cambridge University Press, 1983.
- Hahn, Frank H., "On Non-Walrasian Equilibria," *Review of Economic Studies*, February 1978, 45, 1–18.
- Hart, Oliver, "A Model of Imperfect Competition with Keynesian Features," *Quarterly Journal of Economics*, February 1982, 97, 109–38.
- Heller, Walter P., "Coordination Failure under Complete Markets with Applications to Effective Demand," in *Equilibrium Analysis: Essays in Honor of Kenneth J. Arrow*: Vol. II, Walter P. Heller, Ross M. Starr, and David A. Starrett, eds., Cambridge: Cambridge University Press, 1986, 155–75.

- Ito, Takatoshi**, "An Example of a Non-Walrasian Equilibrium with Stochastic Rationing at the Walrasian Equilibrium Prices," *Economics Letters*, 2, 1979, 13–19.
- Jones, Larry and Manuelli, Rodolfo**, "The Coordination Problem and Equilibrium Theories of Recessions," Kellogg School of Management, Northwestern University, June 1987.
- Kahn, Charles and Mookherjee, Dilip**, "Competitive Efficiency Wage Models with Keynesian Features," Research Paper No. 901, Graduate School of Business, Stanford University, 1986.
- Keynes, John Maynard**, *The General Theory of Employment, Interest and Money*, London: Macmillan, 1936.
- Leijonhufvud, Axel**, *Keynesian Economics and the Economics of Keynes*, London: Oxford University Press, 1968.
- Negishi, Takashi**, *Microeconomic Foundations of Keynesian Macroeconomics*, Amsterdam: North-Holland, 1979.
- Roberts, John**, "General Equilibrium Analysis of Imperfect Competition: An Illustrative Example," in *Arrow and the Ascent of Modern Economic Theory*, George Feiwel, ed., London: Macmillan, 1987, 415–38.
- _____, "Involuntary Unemployment and Imperfect Competition: A Game Theoretic Macro Model," in *The Economics of Imperfect Competition and Employment: Joan Robinson and Beyond*, George Feiwel, ed., London: Macmillan, forthcoming 1988.
- Salop, Steven**, "A Model of the Natural Rate of Unemployment," *American Economic Review*, March 1979, 69, 117–25.
- Selten, Reinhard**, "A Re-examination of the Perfectness Concept for Equilibrium Points in Extensive Games," *International Journal of Game Theory*, 4, 1975, 25–55.
- Shapiro, Carl and Stiglitz, Joseph E.**, "Unemployment Equilibrium as a Worker Discipline Device," *American Economic Review*, June 1984, 74, 433–44.
- Weitzman, Martin**, "Increasing Returns and the Foundations of Unemployment Theory," *Economic Journal*, December 1982, 92, 787–804.